

Netflix Movie Recommendation System using Collaborative Filtering and SVD Matrix Factorization

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Abstract

Recommendation systems have become an essential component of modern digital platforms, helping users discover relevant content from a vast collection of available items. This project focuses on developing a movie recommendation system using the Netflix Prize Dataset. Exploratory Data Analysis (EDA) was performed to understand rating patterns, user activity, and dataset sparsity. Two recommendation techniques were implemented and evaluated: Item-Based Collaborative Filtering (KNN) and Singular Value Decomposition (SVD) Matrix Factorization. The models were trained on a sampled subset of the Netflix dataset consisting of 50,000 ratings from 24,430 users across 504 movies. Performance was evaluated using Root Mean Square Error (RMSE). Experimental results showed that the SVD model achieved better predictive performance than the KNN model. The developed system successfully generated personalized movie recommendations for users based on predicted preferences, demonstrating the effectiveness of recommendation systems in personalized content discovery.

1. Introduction

With the rapid growth of online streaming platforms, users are exposed to an overwhelming number of content choices.

Recommendation systems help users discover movies and shows that match their interests by analyzing historical interactions and rating patterns.

Netflix is one of the most well-known examples of a platform that relies heavily on recommendation systems to improve user experience and engagement. By predicting user preferences, recommendation systems can significantly enhance content discovery and customer satisfaction.

The objective of this project is to develop a movie recommendation system using the Netflix Prize Dataset. The project explores two popular recommendation approaches: Item-Based Collaborative Filtering and Singular Value Decomposition (SVD) Matrix Factorization. The performance of these methods is compared using RMSE, and personalized movie recommendations are generated for users.

2. Dataset Description

The Netflix Prize Dataset is a large-scale collection of movie ratings provided by Netflix for research purposes. It contains ratings assigned by users to movies over a period of time.

To ensure computational efficiency while maintaining sufficient movie coverage, a sampled subset of the dataset was created. Ratings were sampled across multiple movies rather than using a purely sequential subset.

Dataset Statistics:

Metric	Value
Ratings	50,000
Users	24,430
Movies	504

The dataset consists of the following attributes:

- User ID
- Movie ID
- Rating
- Rating Date

These attributes form the user-item interaction matrix used for recommendation generation.

3. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the dataset before model development.

Rating Distribution

The rating distribution indicates that higher ratings occur more frequently than lower ratings. Ratings of 3, 4, and 5 dominate the dataset, suggesting that users generally provide positive feedback for movies they choose to watch.

[Insert Rating Distribution Graph]

User Activity Analysis

User activity is unevenly distributed across the dataset. A small group of highly active users contributes a disproportionately large number of ratings, while most users rate only a few movies.

[Insert User Activity Graph]

Dataset Sparsity

Recommendation datasets are typically sparse because users interact with only a small fraction of available items.

The sparsity of the dataset was calculated using:

$$\text{Sparsity} = 1 - (\text{Number of Ratings} / (\text{Number of Users} \times \text{Number of Movies}))$$

The analysis revealed that the dataset is highly sparse, which presents challenges for recommendation generation and

motivates the use of collaborative filtering and matrix factorization techniques.

[Insert Sparsity Visualization]

EDA Summary

The exploratory analysis revealed three important characteristics:

1. Ratings are skewed toward positive values.
2. User activity is highly uneven.
3. The dataset exhibits high sparsity.

These findings justify the use of recommendation algorithms capable of handling sparse user-item interaction data.

4. Methodology

Two recommendation models were implemented and compared.

4.1 Item-Based Collaborative Filtering (KNN)

Item-Based Collaborative Filtering recommends movies based on similarities between items. The intuition behind this approach is that movies receiving similar rating patterns from users are likely to be related.

Cosine similarity was used to compute similarities between movies. Recommendations were generated by identifying movies similar to those previously rated by a user.

Advantages of Item-Based Collaborative Filtering:

- Easy to interpret
- Widely used recommendation technique
- Effective when sufficient rating data is available

Limitations:

- Performance degrades in highly sparse datasets
 - Computationally expensive for large datasets
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4.2 Singular Value Decomposition (SVD)

Singular Value Decomposition (SVD) is a matrix factorization technique widely used in recommendation systems.

Instead of relying solely on similarity calculations, SVD learns latent factors representing hidden user preferences and movie characteristics. These latent factors enable the model to estimate ratings for unseen user-movie pairs.

Advantages of SVD:

- Handles sparse datasets effectively
- Learns hidden relationships in the data
- Produces more accurate recommendations

Because recommendation datasets are highly sparse, matrix factorization methods often outperform neighborhood-based approaches.

5. Results and Evaluation

Model performance was evaluated using Root Mean Square Error (RMSE).

RMSE measures the difference between actual ratings and predicted ratings. Lower RMSE values indicate better predictive performance.

Model	RMSE
Item-Based Collaborative Filtering (KNN)	1.2377
SVD Matrix Factorization	1.1765

The SVD model achieved a lower RMSE compared to the KNN model, indicating superior predictive performance.

Discussion

The improved performance of SVD suggests that matrix factorization is more effective at capturing hidden relationships between users and movies. Additionally, SVD handles sparse datasets more effectively than traditional neighborhood-based collaborative filtering approaches.

6. Recommendation Generation

After training the SVD model, personalized recommendations were generated for selected users.

The recommendation process involved:

1. Identifying movies not previously rated by a user.
2. Predicting ratings for unseen movies.
3. Ranking movies according to predicted ratings.
4. Recommending the highest-ranked movies.

Sample recommendations generated by the model are shown below.

[Insert Recommendation Table Screenshot]

The generated recommendations demonstrate the ability of the recommendation system to estimate user preferences and suggest relevant unseen content.

7. Conclusion

This project successfully developed a movie recommendation system using the Netflix Prize Dataset.

Major achievements include:

- Performing exploratory data analysis on user rating behavior.
- Analyzing dataset sparsity and user engagement patterns.
- Implementing Item-Based Collaborative Filtering.
- Implementing SVD Matrix Factorization.
- Comparing model performance using RMSE.
- Generating personalized movie recommendations.

Among the evaluated models, SVD achieved the best performance with an RMSE of 1.1765, outperforming the KNN-based collaborative filtering approach.

The results demonstrate the effectiveness of matrix factorization techniques in recommendation systems, particularly when dealing with sparse user-item interaction data.

8. Future Work

Several improvements can be explored in future work:

- Hybrid recommendation systems combining collaborative and content-based approaches.
- Deep learning-based recommendation models.
- Real-time recommendation generation.
- Cold-start problem handling for new users and movies.
- Large-scale deployment using distributed computing frameworks.

These enhancements could further improve recommendation quality and scalability in practical applications.